

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	3.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	22.1500	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	50.0000	50.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	103.6490	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	64.7810	142.5180
Tip diameter (mm)	d_a	70.781 (70.781 / 70.681)	148.518 (148.518 / 148.418)
Root diameter (mm)	d_f	57.281 (57.089 / 56.979)	135.018 (134.757 / 134.619)
Number of teeth spanned	k	3.0000	7.0000
Base tangent length (no backlash) (mm)	W_k	23.186 (23.120 / 23.083)	59.866 (59.777 / 59.730)
Effective Diameter of ball/pin (mm)	D_M	5.2500	5.2500
Diametral measurement over two balls (mm)	M_{dk}	72.217 (72.054 / 71.961)	150.057 (149.818 / 149.692)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.4009		1.4352		1.0000
Bending S.F	4.2091		4.4148		1.4000
Scuffing S.F (Flash Temp.)	2.7044				2.0000
Micropitting S.F	1.0992				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.0000	0.0000	0.0000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4380
Overlap Ratio	2.0002
Total Contact Ratio	3.4382
Worst Pitch Line Velocity [m/s]	33.9192
Total Efficiency [%]	97.6925
Worst Mesh Loss [W]	752.0814
Worst Churning Loss [W]	2709.1082
Worst Peak-to-Peak T.E. [μm]	0.1722
Gear Total Mass [kg]	2.8486
Hunting tooth	Patial

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	3.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	22.1500	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	50.0000	50.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	103.6493	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9667	0.9667
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.2000	
Transverse module (mm)	m_t	3.2390	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.4534	
Working transverse pressure angle (deg)	α_{wt}	21.4534	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.4689 / 21.4379	
Base helix angle (deg)	β_b	20.7502	
Helix angle at operating pitch circle (deg)	β_w	22.1500	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0756 / 0.1188	0.1026 / 0.1566
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0893 / 0.1363
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1695 / 0.2840	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1475 / 0.2471	
Angluar backlash, min (deg)	j_θ, min	0.2998	0.1363
Angluar backlash, max (deg)	j_θ, max	0.5023	0.2283
Effective facewidth (mm)	b_{eff}	50.0000	
Sum of the profile shift coefficients	Sum_x	0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0321 / -0.0504	-0.0435 / -0.0664
Prevent undercut profile shift coefficient	x_{Emin}	-0.4443	-2.1774
Max. profile shift coeff. (min top land)	x_{Emax}	1.1578	1.9903
Reference center distance, $x_1=x_2=0$ (mm)	a_d	103.6493	
Working center distance (mm)	a_w	103.6493	
Reference diameter (mm)	d	64.7808	142.5178
Operating pitch diameter (mm)	d_w	64.7808	142.5178
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	64.7877 / 64.7740	142.5329 / 142.5027
27		70.7808	148.5178
Tip diameter with tolerance (mm)	$d_a, e/i$	70.7808 / 70.6808	148.5178 / 148.4178

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	57.2808	135.0178
Root diameter with tolerance (mm)	d_f, e/i	57.0885 / 56.9786	134.7568 / 134.6194
Tip form diameter (mm)	d_Fa	70.5808	148.3178
Tip form diameter with tolerance (mm)	d_Fa, e/i	70.3631 / 70.2390	148.0390 / 147.8925
Root form diamter (mm)	d_Ff	60.7315	137.3691
Root form diameter with tolerance (mm)	d_Ff, e/i	60.6706 / 60.6379	137.1853 / 137.0899
Active tip diameter (EAP) (mm)	d_Na	70.5808	148.3178
Active tip diameter with tolerance (mm)	d_Na, e/i	70.3631 / 70.2390	148.0390 / 147.8925
Active root diameter (SAP) (mm)	d_Nf	61.0296	138.2932
Active root diameter with tolerance (mm)	d_Nf, e/i	61.1953 / 61.1198	138.4992 / 138.3956
Mean diameter for mass calculation (mm)	d_m	63.8822	141.5780
Gear mass (kg)	m	0.7695	2.0791
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.7500	0.7500
Effective tip clearance, e/i (mm)	c, e/i	1.0102 / 0.8695	0.9621 / 0.8352
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.2246	0.6051

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	6.7500	6.7500
Tooth depth, e/i (mm)	h, e/i	6.9011 / 6.7962	6.9492 / 6.8305
Working depth (mm)	h_w	6.0000	
Addendum (mm)	h_a	3.0000	3.0000
Addendum, e/i (mm)	h_a, e/i	3.0000 / 2.9500	3.0000 / 2.9500
Dedendum (mm)	h_f	3.7500	3.7500
Dedendum, e/i (mm)	h_f, e/i	3.8462 / 3.9011	3.8805 / 3.9492
Number of virtual gear teeth	z_n	24.6932	54.3251
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	5.0879	5.0879
Normal tooth thickness (mm)	s_n	4.7124	4.7124
Normal tooth thickness at d_a (mm)	s_an	2.1643	2.3448
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	2.1424 / 2.0457	2.2914 / 2.1946
Normal tooth thickness at d_Fa (mm)	s_Fan	2.2709	2.4345
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	2.3749 / 2.2674	2.5248 / 2.4086
Normal space width at d_f (mm)	e_fn	0.0000	2.4440
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.4855 / 2.5083

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	9.4248	
Transverse pitch (mm)	p_t	10.1757	
Normal base pitch (mm)	p_bn	8.8564	
Transverse base pitch (mm)	p_bt	9.4707	
Axial pitch (mm)	p_x	24.9972	
Lead (mm)	p_z	499.9449	1099.8788

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_{α}	13.6188	
Length of path of contact, e/i (mm)	g_{α} , e/i	13.1258 / 12.7797	
Length of approach POC of pinion (A to C) (mm)	g_{f1}	7.1183	
Length of approach POC of pinion, e/i (mm)	g_{f1} , e/i	6.8262 / 6.7848	
Length of recess POC of pinion (C to E) (mm)	g_{a1}	6.5005	
Length of recess POC of pinion, e/i (mm)	g_{a1} , e/i	6.2996 / 6.2808	
Length T1 - A (mm)	T1A	4.7283	
T1A tolerance (mm)	e/m/i	5.2364 / 5.1237 / 5.0110	
Length T1 - B (mm)	T1B	8.8764	
T1B tolerance (mm)	e/m/i	8.6661 / 8.6058 / 8.5454	
Length T1 - C (mm)	T1C	11.8466	
T1C tolerance (mm)	e/m/i	11.8560 / 11.8466 / 11.8372	
Length T1 - D (mm)	T1D	14.1991	
T1D tolerance (mm)	e/m/i	14.7071 / 14.5944 / 14.4818	
Length T1 - E (mm)	T1E	18.3471	
T1E tolerance (mm)	e/m/i	18.1368 / 18.0765 / 18.0162	
Length T2 - A (mm)	T2A	33.1809	
T2A tolerance (mm)	e/m/i	32.8681 / 32.7854 / 32.7028	
Length T2 - B (mm)	T2B	29.0328	
T2B tolerance (mm)	e/m/i	29.3938 / 29.3034 / 29.2130	
Length T2 - C (mm)	T2C	26.0626	
T2C tolerance (mm)	e/m/i	26.0832 / 26.0626 / 26.0419	
Length T2 - D (mm)	T2D	23.7101	
T2D tolerance (mm)	e/m/i	23.3973 / 23.3147 / 23.2321	
Length T2 - E (mm)	T2E	19.5621	
T2E tolerance (mm)	e/m/i	19.9231 / 19.8327 / 19.7423	
Length T1 - T2 (mm)	T1T2	37.9092	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	37.9392 / 37.9092 / 37.8791	
Diameter of single contact point A (mm)	d_A	61.0296	148.3178
Diameter of single contact point B (mm)	d_B	62.8518	144.7961
Diameter of single contact point C (mm)	d_C	64.7808	142.5178
Diameter of single contact point D (mm)	d_D	66.6456	140.8651
Diameter of single contact point E (mm)	d_E	70.5808	138.2932
Transverse contact ratio	ϵ_{α}	1.4380	
Overlap ratio	ϵ_{β}	2.0002	
Total contact ratio	ϵ_{γ}	3.4382	
Total contact ratio with tolerance	e/m/i	3.386 / 3.368 / 3.350	
Specific sliding at Tip	ζ_a	0.5154	0.6865
Specific sliding at Root	ζ_f	-2.1898	-1.0634

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	9.0000	9.0000
Base circle pitch deviation (μm)	f_pb	8.4000	8.4000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	16.0000	20.0000
Profile form deviation (μm)	f_fa	9.5000	9.5000
Profile slope deviation (μm)	f_Ha	7.5000	7.5000
Total profile deviation (μm)	F_α	12.0000	12.0000
Helix form deviation (μm)	f_fb	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	10.0000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	26.0000	30.0000
Runout (μm)	F_r	23.0000	27.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	34.5000	38.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	14.0000	15.0000
Total radial composite deviation (μm)	F_idT	36.0000	43.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	3.0000	7.0000
Base tangent length (mm)	W_k	23.1862	59.8661
Base tangent length with tolerance (mm)	W_k, e/i	23.1205 / 23.0829	59.7768 / 59.7299
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	64.0459	143.9326
Theoretical diameter of ball/pin (mm)	D_M.theory	5.1352	5.0672
Effective Diameter of ball/pin (mm)	D_M.eff	5.2500	5.2500
Radial single ball measurement (mm)	M_rk	36.1083	75.0286
Radial single ball measurement with tolerance (mm)	M_rk, e/i	36.0271 / 35.9803	74.9091 / 74.8458
Diametral two ball measurement (mm)	M_dk	72.2167	150.0571
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	72.0542 / 71.9605	149.8181 / 149.6915
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1625 / -0.2561	-0.2390 / -0.3656

3. Detail rating

3.1. General

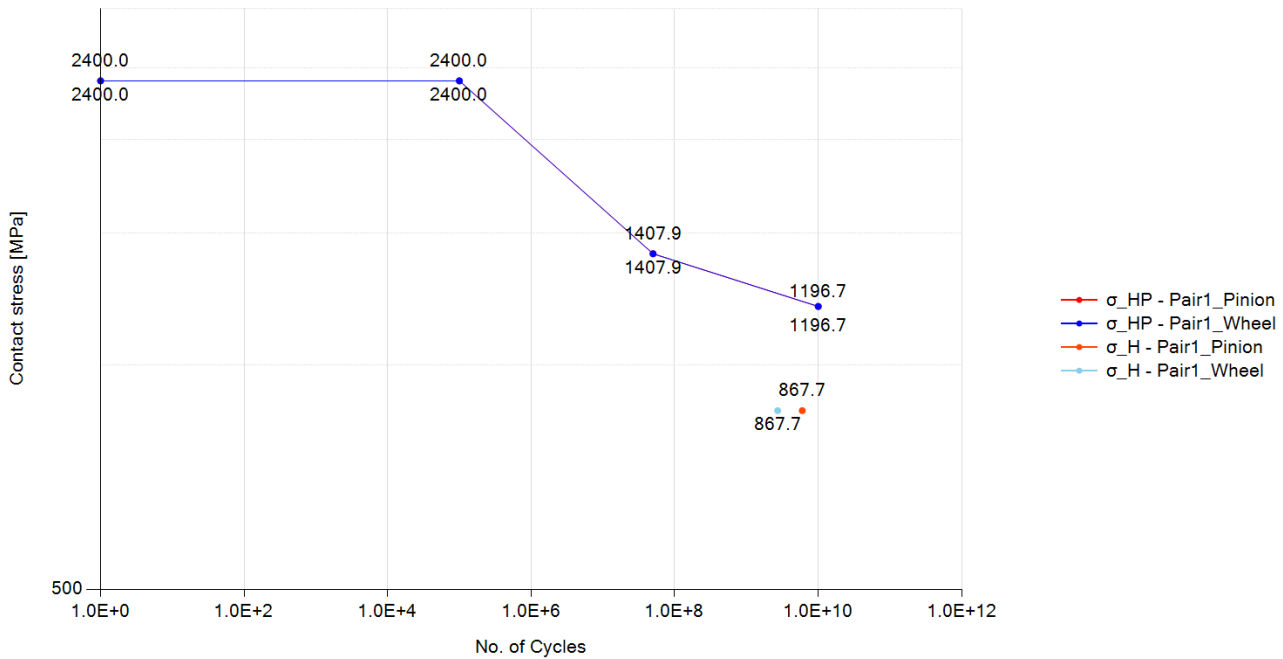
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4545.4550
Nominal tangential load at d (N)	F_t	4422.2790	
Nominal axial load at d (N)	F_a	1800.1980	
Nominal radial load at d (N)	F_r	1737.8300	
Nominal normal load at d (N)	F_n	5081.0750	
Nominal tangential load per unit facewidth (N/mm)	w	88.4460	
Circumferential speed at d (m/s)	v	33.9190	
Sliding velocity at Tip dia. (m/s)	v_g	9.9020	10.8430
Nominal tangential load at d_w (N)	F_{tw}	4422.2790	
Nominal axial load at d_w (N)	F_{aw}	1800.1980	
Nominal radial load at d_w (N)	F_{rw}	1737.8300	
Circumferential speed at d_w (m/s)	v_w	33.9190	
Running-in value (μm)	y_p	0.6300	
Running-in value (μm)	y_f	0.7130	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.7700	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	8.7870	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	17.1600	
Single tooth stiffness (N/mm/ μm)	c'	11.1330	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	14.7890	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	12.5710	
Mean diameter for K_v (mm)	$d_m(K_v)$	64.0310	141.7680
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J(K_v)$	624.7930	10850.3610
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	12.4960	217.0070
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0138	0.0493
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0107	
Resonance running speed (rpm)	n_{E1}	17707.2720	
Resonance ratio	N	0.5650	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_v	1.2610	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2550	
Transverse load factor - flank	$K_{H\alpha}$	1.2450	
Transverse load factor - root	$K_{F\alpha}$	1.2450	
Number of load cycles	N_L	6.0000×10^9	2.7270×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



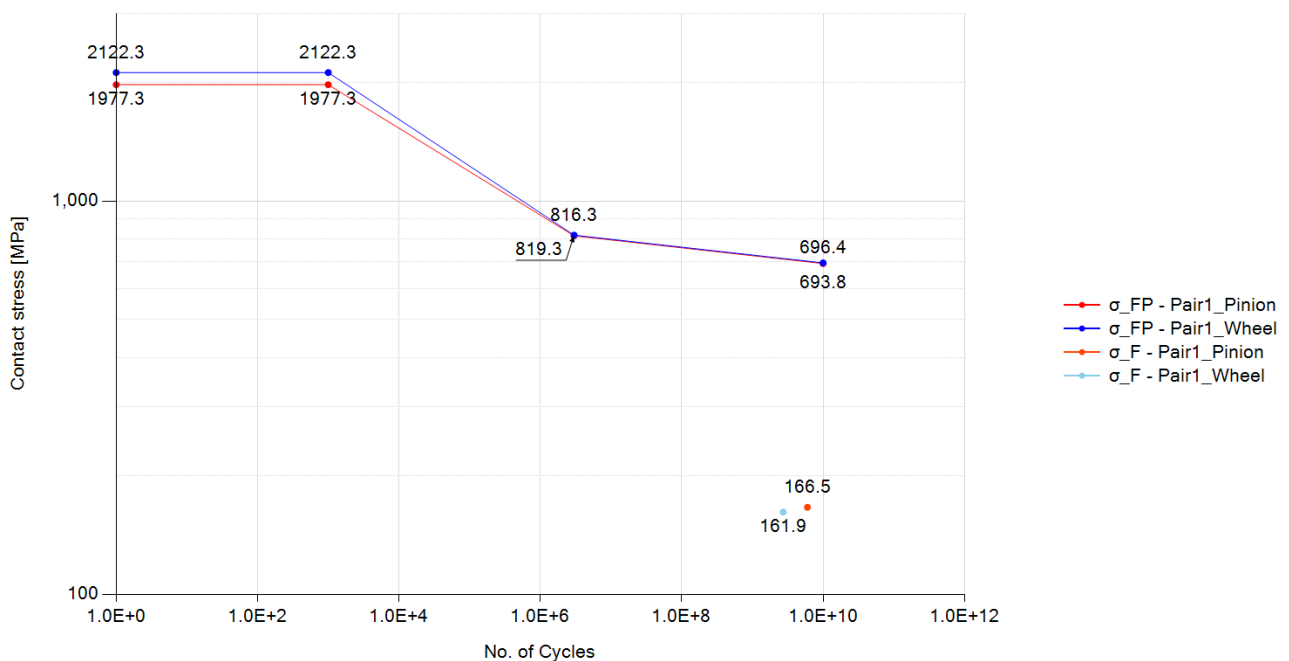
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.8340	
Zone factor	Z_H	2.3440	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0390	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	543.2830	
Contact stress (N/mm ²)	σ_H	867.6880	867.6880
Life factor	Z_{NT}	0.8630	0.8850

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0360	1.0360
Roughness factor	Z_R	0.9580	0.9580
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1407.8670	1407.8670
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1196.6870	1196.6870
Permissible contact stress (N/mm ²)	σ_{HP}	1215.5850	1245.3420
Safety factor for Contact	S_H	1.4010	1.4350
Contact damage (%)	-	0.0020	0.0010
Cycle to failure	-	3.5620×10^{14}	3.5620×10^{14}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2300	1.1030
Stress correction factor	Y_S	1.8170	1.9710

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.5060	19.8370
Bending moment arm (mm)	h_{Fe}	3.0860	3.3140
Tooth thickness at root (mm)	s_{Fn}	5.9610	6.4970
Tooth root radius (mm)	ρ_F	1.7360	1.5760
Helix angle factor	Y_β	1.0260	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	67.6500	65.7650
Tooth root stress (N/mm ²)	σ_F	166.5340	161.8920
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1977.3300	2122.2570
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.2560	819.2510
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	693.8170	696.3640
Permissible tooth root stress (N/mm ²)	σ_{FP}	700.9550	714.7290
Safety factor for Bending	S_F	4.2090	4.4150
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	4.3580×10^{33}	2.1450×10^{34}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	42.5000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	225.6070	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	7.4750	
Angle factor	X_αβ	0.9670	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0420	
Maximum flash temperature (deg)	θ_fl_max	89.0840	
Mean flash temperature (deg)	θ_fl_mean	28.7940	
Total mass temperature (deg)	θ_M	96.2400	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	185.3240	
Safety factor for flash temperature	S_B	2.7040	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.1392
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.2320
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	1.0992
transverse tangential load at reference cylinder per mesh (N)	F_t	4422.2790
transverse load in plane of action (N)	F_bt	4751.4840
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1270
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.3000
load losses factor	H_v	0.1460
mean coefficient of friction	μ_m	0.0480
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.9040
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	20.7460
sliding velocity at point A (m/s)	v_g,A	-10.8430
tangential velocity on pinion at point A (m/s)	v_r1,A	4.9510
tangential velocity on wheel at point A (m/s)	v_r2,A	15.7940
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	96.2920
flash temperature at point A (°C)	θ_fl,A	104.6840
contact temperature at point A (°C)	θ_B,A	200.9760
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3570×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	7.8990×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	1.8890
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0080
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	4.4260
material parameter	G_M	3072.9830
local velocity parameter at point A	U_A	8.2440×10 ⁻¹¹
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.1040
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	836.2920
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1335.6600

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.2000
Tangential speed at working pitch dia. (m/s)	v_wt	33.9192
Tangential force at pitch dia. (N)	F_t	4422.2791
Tooth normal force, transverse section (N)	F_bt	4751.4845
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_Σ	24.8115
Equivalent radius of curvature (mm)	ρ	8.7095
Kinematic viscosity of oil at operating temp. (cSt)	η_oil	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_mz	0.0343
Tooth loss factor	H_v	0.1461
Mesh power loss (W)	P_VZP	752.0814
Mesh efficiency (%)	η_z	99.4986

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4545.4545
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	32.3904	71.2589
Oil density (kg/m^3)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	1.1413×10^5	1.1413×10^5
Froude number	Fr	3620.8016	1645.8189
Centrifugal acceleration (m/s^2)	γ	18573.4745	4991.0128

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	70.7808	148.5178
Pitch diameter of gear (mm)	d_p	64.7808	142.5178
Facewidth (mm)	b	50.0000	50.0000
Dimensionless drag torque	C_m	0.0013	0.0031
Immersion area of the gear (m ²)	S_m	0.0161	0.0442
Total oil churning loss (W)	P	365.1851	2343.9231
Total oil churning torque (N.m)	T	0.3487	4.9242

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000